

EnPoWER Insti-Assist — Project Write-Up

A RAG-powered assistant grounded in EnPoWER's council documents and the IIT Bombay Students' Gymkhana Constitution

1. Chosen Scope and Why

We built a **Council/Club Assistant** for **EnPoWER** — the Engineering Oriented Promotion of Work Experience and Research initiative, IIT Bombay's undergraduate research-promotion council. This scope was chosen over a general insti-wide assistant for two reasons. First, EnPoWER's own documents (work reports, selection assignments) are dense with specific, non-obvious facts — programme names, dates, coordinator structures — that are exactly the kind of thing a general LLM would hallucinate rather than know. Second, EnPoWER's activities are formally anchored in the Students' Gymkhana Constitution, so the corpus naturally spans two registers: operational reporting (what the council actually did) and constitutional/procedural text (what it is allowed and required to do). That mix stress-tests chunking and retrieval more than a single homogeneous document type would.

2. Data Sources Used

Seven source documents (two above the required minimum of five), all real EnPoWER / IIT Bombay documents:

- End-Term Work Report, 2023-24 (PDF)
- End-Term Work Report, 2024-25 (DOCX)
- End-Term Work Report, 2025-26 — Final (PDF)
- Mid-Term Work Report, 2025-26 (PDF)
- Academic Coordinator selection assignment, Round 2 / UGAC (PDF)
- Department Research Coordinator (DRC) selection assignment, 2026-27 (DOCX)
- IIT Bombay Students' Gymkhana Constitution, March 2018 (PDF, 81 pages) — the governance document EnPoWER and all other councils operate under

3. Chunking Strategy and Why

Extraction is provenance-first: PDFs are split page-by-page with pdfplumber, and DOCX files are split by Word heading styles (falling back to one block if a document has no headings, as with the DRC assignment), plus a separate pass that pulls table rows out as their own units. Every extracted unit carries its exact source file name and page/section label from the start, rather than trying to reconstruct that mapping later.

On top of that, a **paragraph-packing chunker** greedily merges paragraphs into ~900-character windows (roughly 200-250 tokens) with a 150-character overlap between consecutive chunks from the same page/section. Three deliberate choices here: (1) we chunk on paragraph boundaries, never mid-sentence, so a chunk always reads as a coherent unit; (2) we never merge text across two different pages/sections into a single chunk — this costs some packing efficiency but keeps every chunk's source citation exact instead of an approximate range; (3) the overlap prevents an answer that happens to sit right on a chunk boundary from being split across two chunks and under-retrieved. 900 characters was chosen empirically as a middle ground: work-report paragraphs (bullet-heavy, specific) stay mostly intact in one chunk, while the constitution's denser clauses (occasionally >1,350 characters) fall back to a hard character-split safety net.

4. Known Limitations / What We'd Improve With More Time

- **No labeled eval set.** The similarity gate (cosine ≥ 0.30 on the top retrieved chunk, below which the assistant refuses without calling the LLM) was set by manual inspection of a handful of queries, not tuned against a labeled set of in-scope vs. out-of-scope questions. A proper eval — e.g. 30-50 questions labeled answerable/unanswerable — would let us tune this threshold and measure precision/recall on refusals directly instead of eyeballing it.
- **No OCR fallback.** All seven source PDFs are text-native, so this wasn't an issue here, but a scanned or image-only page would silently extract as empty text today.
- **DOCX section granularity is uneven.** The 2024-25 end-term report is well headinged and chunks cleanly; the DRC assignment doc has no heading styles, so its citation label falls back to a generic "Document start" tag rather than a specific section name.
- **Single retrieval pass, no re-ranking.** We use flat cosine similarity over all-MiniLM-L6-v2 embeddings with no cross-encoder re-ranking step. For this corpus size (~400 chunks) that's adequate, but a cross-encoder re-rank on the top-20 candidates would likely improve precision on ambiguous queries (e.g. questions that could refer to either the 2024-25 or 2025-26 report).
- **No multi-turn memory yet.** Each query is currently answered independently; follow-up questions like "and what about the year before?" aren't resolved against conversation history. This was our first-choice bonus item to add next.
- **Single embedding model, no hybrid search.** Dense retrieval alone can miss exact keyword/acronym matches (e.g. "SIG", "DRC", "ISAA"); adding a sparse BM25/keyword pass alongside the dense one and combining scores would likely help on acronym-heavy queries specifically.